

HEMANTH KUMAR CHITTIPROLU

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PROFESSIONAL SUMMARY

AI/ML Engineer with hands-on experience building and deploying production AI systems across NLP, computer vision, and generative AI. Skilled in end-to-end ML pipelines, gradient boosting models, LLM applications (RAG, fine-tuning), web scraping at scale, and cloud deployment on AWS and Azure. Proven track record of solving real-world problems with noisy data, advanced feature engineering, and Bayesian methods. Passionate problem solver who thrives on ambiguity — bringing the engineering judgment, domain intuition, and persistence needed to ship real systems where off-the-shelf AI falls short.

TECHNICAL SKILLS

Programming Languages: Python, C++, SQL, JavaScript

ML / Deep Learning: Scikit-learn, TensorFlow, Keras, PyTorch, LightGBM, XGBoost, Gradient Boosting, CNN, RNN, LSTM, GANs, Transfer Learning

NLP: Hugging Face Transformers, BERT, GPT, SpaCy, NLTK, Tokenization, Text Classification, Sentiment Analysis, Summarization

Computer Vision: OpenCV, YOLO, Image Classification, Object Detection, Image Segmentation

Generative AI / LLMs: LangChain, LlamaIndex, RAG (Retrieval-Augmented Generation), PEFT, LoRA, Prompt Engineering, Vector Databases (Qdrant, Pinecone, FAISS), Nomic Embeddings

MLOps & Deployment: MLflow, FastAPI, Docker, Kubernetes, CI/CD, REST APIs, Model Monitoring, Hyperparameter Tuning (Optuna, GridSearch, RandomSearch), Cross-Validation

Data Engineering: Pandas, NumPy, Feature Engineering, Data Preprocessing, Stratified Sampling, PCA, t-SNE, ETL Pipelines

Statistics: Hypothesis Testing, Regression Analysis, Time Series, Bayesian Statistics, A/B Testing

Cloud Platforms: AWS (S3, Athena, EC2), Microsoft Azure (Databricks, Azure ML)

Web Scraping: Playwright, Puppeteer, Camoufox, Scrapy, Selenium, Anti-Bot Bypass, Proxy Rotation

Web Development: FastAPI, React.js, HTML, CSS, JavaScript, Node.js

Tools & Platforms: Git, GitHub, Jupyter, Google Colab, Anaconda, Linux, Apify, Power BI, Matplotlib, Seaborn, Plotly

WORK EXPERIENCE

AI/ML Engineer — Knackhook

June 2025 – Present | Hyderabad, India

- Architected and deployed production AI systems across NLP pipelines, computer vision applications, and cloud deployments on Microsoft Azure.
- Designed and built an end-to-end ML data pipeline for ticket price prediction integrating AWS S3 and AWS Athena, including stratified sampling and data cleaning of large-scale Excel-based source data.
- Developed a semantic code search system for an internal codebase using Nomic embeddings (deployed via Modal) and Qdrant vector database, improving developer search efficiency.
- Built and deployed a high-volume web scraper bypassing Akamai bot detection on Ticketmaster, deployed on Apify for scalable production data collection.
- Researched and prototyped hallucination detection methodologies using BERT-based stochastic checkers for production LLM systems.
- Contributed to KnackCampaigner (email marketing automation), KnackHire (recruitment platform), and KnackOutreach (outreach automation) products as part of the AI engineering team.

PROJECTS

Ticket Price Prediction System

- Built an end-to-end ML pipeline to predict ticket prices using LightGBM and Gradient Boosting models.
- Handled noisy and incomplete real-world event data through robust preprocessing, outlier treatment, and missing-value imputation.
- Applied advanced feature engineering including temporal features, interaction terms, and high-cardinality categorical encoding.
- Used Bayesian statistics to complement model predictions and quantify confidence in outputs.
- Integrated AWS S3 for data storage and AWS Athena for scalable querying as part of the data pipeline.

Anti-Bot Web Scraper – Ticketmaster & High-Security Platforms

- Built a stealth scraper targeting Ticketmaster and similar high-security ticketing platforms protected by Akamai bot detection.
- Bypassed advanced anti-bot mechanisms using stealth browser automation (Playwright, Puppeteer, Camoufox), browser fingerprint spoofing, and rotating proxy strategies.
- Deployed on Apify cloud infrastructure for scalable, production-grade data collection.
- Collected structured event and pricing data used as input for the ticket price prediction pipeline.

RAG-Based Academic Research Assistant

- Designed and implemented a Retrieval-Augmented Generation (RAG) system for academic literature search and Q&A.
- Integrated vector database storage, semantic chunking, and LLM-based response generation with citation grounding.

LLM Evaluation Framework for Multi-Domain Applications

- Developed an evaluation framework to benchmark LLM performance across multiple domains using accuracy, hallucination rate, and task-specific metrics.
- Automated batch evaluation pipelines using Hugging Face datasets and custom scoring functions.

LLM-Powered Android App Security Tester

- Built an automated security testing tool that uses LLMs to analyze Android applications for vulnerabilities.
- Combined static analysis with LLM-driven reasoning to identify potential security issues.

Sales Forecasting & Time Series Models

- Developed time-series forecasting models using statistical methods and ML approaches for retail sales prediction.

Diabetes Prediction with PySpark MLlib

- Implemented distributed ML classification on the Pima Indians dataset using Spark MLlib for scalable training.

Medical Insurance Premium Prediction

- Built regression models to predict insurance premiums based on demographic and health features; performed feature importance analysis.

IoT-Based Smart Agriculture System

- Developed an IoT-driven precision agriculture platform with real-time soil and environmental sensor data.
- Applied ML algorithms to provide actionable insights for crop monitoring and yield optimization.

EDUCATION

B.Tech in Artificial Intelligence & Machine Learning

Anurag University, Hyderabad | 2020 – 2024

CERTIFICATIONS

- IBM AI Engineering Professional Certificate

- Google Data Analytics Professional Certificate
- Google Business Intelligence Professional Certificate
- Google Cybersecurity Professional Certificate
- Google Project Management Professional Certificate
- Microsoft Power BI Data Analyst Professional Certificate
- Machine Learning for Trading Specialization (Coursera)
- Unilever Supply Chain Data Analyst Professional Certificate
- IBM Penetration Testing, Incident Response and Forensics
- Cisco — Introduction to Cybersecurity & Cybersecurity Essentials
- Infosys Springboard — Python and Java Certifications

ACHIEVEMENTS & EXTRACURRICULAR

- 1st Position — IIT Hyderabad Inspira Hackathon.
- Special Mention — Reboot the Earth Hackathon by United Nations at Salesforce.
- 4th Position — Google Hackathon at ISB Hyderabad (supported by T-Hub), evaluated by Google employees and investors.
- 4th Position — GreenBiz Hackathon, Narayanamma College.
- Winner — Ideathon at Osmania University.
- Organized the WIE Conference for IEEE SNIST at The Park, Hyderabad.
- Coordinated a Web 3.0 workshop and led a robotic racing event at college level.